

Estimated Shielded-Output Crypto Times Across Devices

Hathor Network — Issue #1603 PoC

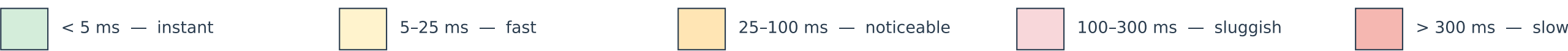
Methodology

Times are extrapolated from a measured baseline on Intel Core i5-11300H (single-threaded, WSL2 Linux 6.6) by scaling with the ratio of Geekbench 6 single-core scores. The baseline is the diagonal of the full per-output workload benchmark (results_full/total_create.csv and results_full/total_verify.csv). Formula: $\text{Time}(\text{device}) = \text{Time}(\text{i5-11300H}) \times (1900 / \text{GB6_single_core_score}(\text{device}))$.

Caveats

Real on-device performance can swing $\pm 30\text{--}50\%$ from this estimate. The crypto stack benefits from SIMD (AVX2 on x86, NEON on ARM) and from secp256k1 hand-tuned assembly; mobile chips also throttle under sustained load. Treat values as order-of-magnitude. GB6 scores for 2014 devices are extrapolated from Geekbench 4/5 since GB6 does not run on those iOS/Android versions.

Color legend (per cell)



Estimated TOTAL CREATION Time per Transaction

Pedersen commitment + Bulletproof range proof + asset surjection proof, summed across all M outputs.

Device	Category	×	(N=1, M=1)	(N=2, M=2)	(N=4, M=4)	(N=8, M=8)	(N=16, M=16)	(N=32, M=32)	(N=64, M=64)
Intel Core i5-11300H (reference)	Reference (this PoC)	1.00×	1.35 ms	2.64 ms	7.10 ms	11.1 ms	24.8 ms	47.7 ms	112 ms
iPhone 16 Pro Max	Most powerful	0.56×	757 μs	1.47 ms	3.97 ms	6.20 ms	13.9 ms	26.6 ms	62.7 ms
Samsung Galaxy S25 Ultra	Most powerful	0.63×	858 μs	1.67 ms	4.49 ms	7.03 ms	15.7 ms	30.2 ms	71.1 ms
OnePlus 13	Most powerful	0.63×	858 μs	1.67 ms	4.49 ms	7.03 ms	15.7 ms	30.2 ms	71.1 ms
iPhone 15	USA top-3	0.76×	1.03 ms	2.00 ms	5.39 ms	8.44 ms	18.8 ms	36.2 ms	85.3 ms
iPhone 14	USA + Europe top-3	0.83×	1.12 ms	2.18 ms	5.86 ms	9.17 ms	20.5 ms	39.4 ms	92.8 ms
Samsung Galaxy S23	USA top-3	0.95×	1.29 ms	2.51 ms	6.74 ms	10.5 ms	23.6 ms	45.3 ms	107 ms
Samsung Galaxy A54	Europe + ROW top-3	1.81×	2.45 ms	4.77 ms	12.8 ms	20.1 ms	44.9 ms	86.3 ms	203 ms
Xiaomi Redmi Note 12	Europe + ROW top-3	2.07×	2.80 ms	5.45 ms	14.7 ms	22.9 ms	51.2 ms	98.5 ms	232 ms
Samsung Galaxy A14	Rest of World top-3	3.96×	5.36 ms	10.4 ms	28.1 ms	43.9 ms	98.1 ms	189 ms	444 ms
Tecno Spark 10	Rest of World top-3	3.80×	5.15 ms	10.0 ms	27.0 ms	42.2 ms	94.2 ms	181 ms	427 ms
iPhone 6 (2014)	2014 era	5.43×	7.35 ms	14.3 ms	38.5 ms	60.3 ms	135 ms	259 ms	610 ms
Samsung Galaxy S5 (2014)	2014 era	6.79×	9.19 ms	17.9 ms	48.2 ms	75.3 ms	168 ms	324 ms	762 ms
LG G3 (2014)	2014 era	6.79×	9.19 ms	17.9 ms	48.2 ms	75.3 ms	168 ms	324 ms	762 ms

Estimated TOTAL VERIFICATION Time per Transaction

Pedersen commitment + range proof + surjection verification, summed across all M outputs.

Device	Category	×	(N=1, M=1)	(N=2, M=2)	(N=4, M=4)	(N=8, M=8)	(N=16, M=16)	(N=32, M=32)	(N=64, M=64)
Intel Core i5-11300H (reference)	Reference (this PoC)	1.00×	819 μs	1.82 ms	4.69 ms	7.31 ms	16.7 ms	34.1 ms	82.6 ms
iPhone 16 Pro Max	Most powerful	0.56×	458 μs	1.02 ms	2.62 ms	4.08 ms	9.32 ms	19.1 ms	46.1 ms
Samsung Galaxy S25 Ultra	Most powerful	0.63×	519 μs	1.15 ms	2.97 ms	4.63 ms	10.6 ms	21.6 ms	52.3 ms
OnePlus 13	Most powerful	0.63×	519 μs	1.15 ms	2.97 ms	4.63 ms	10.6 ms	21.6 ms	52.3 ms
iPhone 15	USA top-3	0.76×	622 μs	1.38 ms	3.56 ms	5.55 ms	12.7 ms	25.9 ms	62.7 ms
iPhone 14	USA + Europe top-3	0.83×	677 μs	1.50 ms	3.87 ms	6.04 ms	13.8 ms	28.2 ms	68.2 ms
Samsung Galaxy S23	USA top-3	0.95×	778 μs	1.73 ms	4.45 ms	6.94 ms	15.8 ms	32.4 ms	78.4 ms
Samsung Galaxy A54	Europe + ROW top-3	1.81×	1.48 ms	3.29 ms	8.48 ms	13.2 ms	30.2 ms	61.8 ms	149 ms
Xiaomi Redmi Note 12	Europe + ROW top-3	2.07×	1.69 ms	3.76 ms	9.68 ms	15.1 ms	34.5 ms	70.5 ms	170 ms
Samsung Galaxy A14	Rest of World top-3	3.96×	3.24 ms	7.20 ms	18.6 ms	28.9 ms	66.0 ms	135 ms	327 ms
Tecno Spark 10	Rest of World top-3	3.80×	3.11 ms	6.92 ms	17.8 ms	27.8 ms	63.4 ms	130 ms	314 ms
iPhone 6 (2014)	2014 era	5.43×	4.45 ms	9.88 ms	25.5 ms	39.7 ms	90.6 ms	185 ms	448 ms
Samsung Galaxy S5 (2014)	2014 era	6.79×	5.56 ms	12.3 ms	31.8 ms	49.6 ms	113 ms	232 ms	560 ms
LG G3 (2014)	2014 era	6.79×	5.56 ms	12.3 ms	31.8 ms	49.6 ms	113 ms	232 ms	560 ms

Sources & References

Geekbench 6 single-core scores

Primary database: <https://browser.geekbench.com/> (search by device or chipset)
Mac/iOS chart: <https://browser.geekbench.com/mac-benchmarks>
Android chart: <https://browser.geekbench.com/android-benchmarks>
Processor chart: <https://browser.geekbench.com/processor-benchmarks>
2014 devices (iPhone 6, Galaxy S5, LG G3) extrapolated from Geekbench 4/5 archives, since GB6 requires iOS 12+ / modern Android.

Device specifications (SoC, model details)

GSMarena device database: <https://www.gsmarena.com/>
Apple newsroom & technical specs: <https://www.apple.com/newsroom/> / <https://support.apple.com/specs>
Samsung official spec pages: <https://www.samsung.com/global/galaxy/>
Qualcomm Snapdragon spec sheets: <https://www.qualcomm.com/products/mobile/snapdragon>
MediaTek SoC pages: <https://www.mediatek.com/products/smartphones>

Device popularity / market share

USA top sellers (iPhone 15, iPhone 14, Galaxy S23):
- Counterpoint Research US smartphone tracker: <https://www.counterpointresearch.com/>
- CIRP US iPhone reports: <https://cirpllc.com/>
Europe top sellers (iPhone 14, Galaxy A54, Redmi Note 12):
- Canalys EMEA quarterly: <https://canalys.com/newsroom>
- IDC Western Europe Mobile Phone Tracker: <https://www.idc.com/>
Rest of World (Samsung Galaxy A14, Tecno Spark 10, Redmi Note 12):
- IDC Worldwide Quarterly Mobile Phone Tracker: <https://www.idc.com/>
- Canalys Africa & Latin America reports: <https://canalys.com/>
- StatCounter GlobalStats vendor share: <https://gs.statcounter.com/vendor-market-share/mobile>

Most powerful (2025) selection rationale

iPhone 16 Pro Max (A18 Pro), Samsung Galaxy S25 Ultra (Snapdragon 8 Elite), OnePlus 13 (Snapdragon 8 Elite) selected as the top three single-core performers on the Geekbench 6 mobile leaderboard at time of writing (April 2026).

Disclaimer

Market-share rankings shift quarter to quarter. The devices chosen here represent the top of recent reports (2024–2025) and are intended as illustrative anchors, not as a definitive ranking. Multipliers in the table are derived solely from Geekbench 6 single-core scores; on-device benchmark of the actual hathor-ct-crypto library on each phone would be required for production-grade numbers.